MILITARY SPECIFICATION

INTRAUTERINE DEVICE, CONTRACEPTIVE, PLASTIC, RADIOPAQUE, 30 mm WIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of sterile, radiopaque, intrauterine device and sterile insertion unit suitable for contraceptive purposes.

1.2 Classification. Intrauterine devices shall be furnished in the following sizes as specified (see 6.1):

Size	Mass
C	approximately 615 milligrams (mg)
D	approximately 709 milligrams (mg)

2, APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	- Adhesive, Water-Resistant (for Sealing fiberboard boxes).
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-T-76	- Tape, Pressure Sensitive Adhesive Paper (for carton sealing).

Military

MIL-S-36586	- Sterilization Strip Test, Bacterial Spore.

FSC 6515

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of American (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 19042).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall be as indicated herein and of the highest quality suitable in all respects for the purpose intended.

3.1.1 Loop. The body portion of the loop shall be fabricated from nontoxic, flexible, low-density polyethylene. The tail thread portion of the loop shall be fabricated from nontoxic, flexible, linear polyethylene-monofilament. The plastic loop shall meet the requirements of the United States Pharmacopeia (USP) for class VI plastic, when tested as specified in 4.5.1.

3.1.2 Inserter. The insertion unit (sleeve and plunger) shall be fabricated from nontoxic, flexible plastic of the best quality and suitable in all respects for the purpose intended.

3.1.3 Barium sulfate. The intrauterine loop shall contain USP grade barium sulfate to render the device visible to X-ray examination. The body portion of the loop shall contain 25.7 to 31.4 percent by volume of USP grade barium sulfate, when tested in accordance with 4.5.3.

3.1.4 Gloves. Gloves, when supplied, shall be fabricated from the best quality material and suitable in all respects for the purpose intended.

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3.2 Style, design, and dimensions. The intrauterine device shall be a sterile, radiopaque, resilient loop suitable for contraceptive purposes. The loop shall consist of an elongated body and a thread-like tail. The body portion shall be curved in a substantially flat plane in the shape of a large "S" continuing into a smaller "S" to form a continuous double "S" curved shape and shall have a nominal length of 5-1/8 inches when straightened out. The width of the device, measured across the top of the large "S", shall be a nominal 30 millimeters (30 mm.). The tail thread shall be securely attached, at one end, to the tip of the small "S" and, at the other end, to the distal end of the inserter plunger. The thread shall have a length of 3-7/8 to 4-1/4 inches and shall be colored yellow for size C and white for size D. Size C loops shall weigh approximately 615 mg. and shall have reduced radii in the sharply curved portions of the body. Size D loops shall weigh approximately 709 mg. and shall have uniform cross sections throughout the entire body of the loop. The intrauterine loop shall be provided with an insertion unit which shall consist of a tubular guide sleeve and a rod or plunger. The plunger shall have sufficient clearance between its periphery and the internal surface of the guide sleeve to accommodate the tail thread and to make a free, sliding fit within the sleeve. There shall be a flat, oval-shaped flange located approximately 4.4 centimeters (cm.) from the distal end of the tube to serve as an indicator for insertion. The cross section of the plunger shall conform to that of the tube so that, when the plunger is inserted into the tube, the flange at the proximal end of the plunger is in the same plane as the indicator flange on the tube. The distal end of the loop shall not pass through the inserter tube when the loop is drawn into the tube. The intrauterine device may be furnished with a pair of sterile gloves.

3.3 Sterility. The loop, insertion unit, and gloves (if furnished) shall be sterile when tested with a biological indicator as specified in 4.5.2.

3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item Identification.
b. National Stock Number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.

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g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility (lot) control number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.4 Animal implantation. When tested as specified in 4.5.1.1, the intrauterine devices shall show no evidence of toxicity or adverse tissue reaction upon gross and microscopic examination.

3.5 Certificates of biological tests and animal implantation tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested in accordance with methods described in the USP and found to have passed the biological tests and the animal implantation test. This certificate shall include the specific tests performed and the test results. The record of the animal implantation test and its result shall be certified by the pathologist who performed or supervised performance of the animal implantation test. For destination inspection, certificates shall be submitted by the contractor directly to the Defense Personnel Support Center, Directorate of Medical Materiel, 2800 South 20th Street, Philadelphia, PA 19101, attention: Quality Assurance Branch and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

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3.6 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all sterility, biological, and animal implantation tests performed on the intrauterine devices. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.7 Instructions. Each intrauterine device shall be provided with a detailed instruction sheet containing the following minimum information: description of the device; indications and contraindications for use; time of insertion; techniques for correct insertion and removal of the device with appropriate anatomy illustrations; possible side effects; and patient instructions. All instructions shall be in accordance with those of Food and Drug Administration.

3.8 Labeling. Physician and patient product labeling shall provide information on the nature of the IUD, its side effects, contraindications, and potential hazards, and the need for continued medical supervision in accordance with those of Food and Drug Administration.

3.9 Workmanship. Workmanship shall be first class throughout. The intrauterine devices shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic or the barium sulfate, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one intrauterine device and inserter.

TABLE I. Sampling for examination

Defect category	Inspection level	AQL (percent defective)
Major	S-2	1.0
Minor	S-1	1.5

4.2.2 For tests.

4.2.2.1 For sterility test. Sampling for the sterility test shall be conducted as indicated in the **USP for non-parenteral** devices.

4.2.2.2 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The unit of product shall be one intrauterine device and inserter.

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TABLE II. Sampling criteria for tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Unit of product	Material-class VI	3.1.1	4.5.1	2/	1/
Unit of product	Dimensions	3.2	-----	S-2	1.0
Unit of product	Sterility	3.3	4.5.2	S-2	1/
Unit of product	Animal Implantation	3.4	4.5.1.1	S-1	1/
Inserter	Dimensions	3.2	------	S-2	1.0
IUD	Barium Sulfate	3.1.3	4.5.3	S-2	1.0

1/ Acceptance number shall be zero.

2/ Sufficient samples to perform the biological tests shall be selected from each lot and shall be representative of each batch of plastic used in the fabrication of the loops.

4.2.3 Raw materials. Sufficient quantity of each lot of barium sulfate and each lot of each plastic used in the construction of the loops shall be selected for chemical analysis.

4.3 Examination. The intrauterine loops and inserters shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in tables I and III. Examination, however, shall not be restricted to the classified possible defects listed in table III. Any intrauterine device and inserter in the sample containing one or more defects shall be rejected. If the number of defective devices in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	One or more components missing.
102	Loop, guide sleeve, or plunger not free of cracks, burrs, and slivers.
103	Surfaces of loop, plunger, or guide sleeve not smoothly finished, and not free of flash, and feathering.

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TABLE III. Classification of defects (continued)

Categories	Defects
Major	
104	Guide sleeve lumen not free of obstructing material, burs, chips, dirt, and other foreign material.
105	Plunger does not make a free, sliding fit within guide sleeve.
106	Instruction sheet missing, incomplete, or illegible.
Minor	
201	Tail thread not appropriately color coded.

4.4 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE IV. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging & packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Marking	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Biological tests for class VI plastic. The loop shall be tested in accordance with the requirements of the USP for class VI plastic.

4.5.1.1 Animal implantation test. Samples of the finished loop (end item) shall be tested in accordance with the USP for Implantation Test, Plastic Containers, for class VI plastic, except after "72 hours" add "and not to exceed 144 hours"; and delete "Examine macroscopically the area of the tissue surrounding the center portion of each implant strip. Use a magnifying lens if necessary" and substitute: "The muscle tissue surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American College of Veterinary Patholgists."

4.5.2 Sterility test. The intrauterine devices, insertion units, and gloves (when supplied) shall be tested for sterility in accordance with the method specified in the USP for non-parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.5.2.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table II, Sampling criteria for tests.

4.5.2.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of sterilization	Organism	Incubation temperature
Steam	B. stearothermophilus	55° - 65°C
Dry heat	B. stearothermophilus	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.5.2.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.5.2 Sterility test and 4.5.2.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.5.2.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.5.2.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Request for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.5.2.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P.

4.5.3 Barium sulfate. Accurately weigh a 6.5 gram sample of the IUD plastic and proceed in accordance with the USP method of assay for barium sulfate.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each intrauterine device along with an inserter and instructions for use shall be mounted on a paperboard stiffener or other suitable material. The complete unit shall be packaged sterile in a suitable sealed peel open type bag of an appropriate size and design. The bag shall be designed with tabs to facilitate opening and in addition, the bag shall be capable of maintaining sterility of contents unless the bag is damaged or opened.

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5.1.1.2 Intermediate package. Twenty-four unit packages containing intrauterine devices of one size only shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, type CF, class weather-resistant. The box shall be closed by taping the top and bottom flaps with a 2 inch wide tape conforming to PPP-T-76. Closing of the bottom with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four intermediate packages (96 units) containing intrauterine devices of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. Four intermediate packages (96 units) containing intrauterine devices of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall conform to method I or II as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. In addition to any commercial markings, each unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall also include the sterility lot (control) number, contract or purchase order number and name of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold,

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S-Steam, D-Dry Heat) shall be shown following the level of packaging and date of packaging. Example: "A-8/75-G". Final sterilization shall be on or after indicated packaging date. The legend "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement shall also be shown. Marking shall be printed directly on the bag or on a label inserted within the bag in such a manner as to permit ready identification.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, contract or purchase order number, name of contractor and the method of sterilization as specified in 5.3.1 shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number and the method of sterilization as specified in 5.3.1 shall be shown.

6. NOTES

6.1 Intended use. The intrauterine devices are intended for optimum uterine placement, fit, tolerance and retention for effective contraception.

6.2 Ordering data. Purchasers should exercise any desired options specified herein, and the procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Classification (see 1.2).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

6.3 National Stock Number (NSN). This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification designation Size
6515-00-145-0401	INTRAUTERINE DEVICE, Contraceptive, Plastic, Radiopaque, 30mm Wide	Size C, 615 mg

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National Stock Number	Item Identification	Specification designation Size
6515-00-483-7987	INTRAUTERINE DEVICE, Contraceptive Plastic, Radiopaque, 30mm Wide	Size D, 709 mg

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1969
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-I-37278 INTRAUTERINE DEVICE, CONTRACEPTIVE, PLASTIC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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